

Q1. Explain Cryptography and Discuss its Goals in Securing Digital Communication.

- **Cryptography** is the science of securing information by converting readable data (**plaintext**) into an unreadable form (**ciphertext**) using an encryption algorithm and a secret key. Only an authorized user having the correct key can decrypt the ciphertext back into the original message. It is one of the most important techniques used to protect digital communication from unauthorized access, hacking, and data theft.

Example:

Plaintext: HELLO

Encrypted (Ciphertext): KHOOR

After decryption, the receiver gets the original message **HELLO**.

Goals of Cryptography

1. **Confidentiality** : Confidentiality ensures that only authorized users can read the information. Encryption protects sensitive data from unauthorized persons.
Example: Internet banking, online shopping, and WhatsApp messages
2. **Integrity** : Integrity ensures that the data is not modified during transmission. If someone changes the message, the receiver can detect the modification.
Example: If the amount ₹1,000 is changed to ₹10,000, cryptographic techniques can detect the alteration.
3. **Authentication** : Authentication verifies the identity of the sender and receiver. It confirms that the message is sent by a genuine user.
Example: Logging into Gmail using a password and OTP verification in online banking.
4. **Non-Repudiation** : Non-repudiation ensures that the sender cannot deny sending a message after it has been transmitted. This is achieved using **Digital Signatures**.
5. **Data Privacy** : Cryptography protects personal and confidential information from unauthorized access and cyber-attacks.

Applications of Cryptography

1. **Online Banking** – Secures banking transactions and protects customer account details.
2. **Email Security** – Encrypts emails to prevent unauthorized access.
3. **Digital Payments** – Protects UPI, credit/debit card, and online payment transactions.

Advantages of Cryptography

1. Provides Data Confidentiality
2. Maintains Data Integrity
3. Ensures Secure Communication

Q2. Describe the CIA Triad with Suitable Real-World Examples.

- The **CIA Triad** is the basic security model used in Cyber Security and Cryptography. It helps protect information from unauthorized access, modification, and loss. The three main principles of the CIA Triad are:
 - **C – Confidentiality**
 - **I – Integrity**
 - **A – Availability**
- These principles ensure that information remains secure and accessible only to authorized users.

1. Confidentiality: Confidentiality means that information should be accessible only to authorized users. It prevents unauthorized people from viewing sensitive data.

Example: Only the account holder can view their online banking details using a username and password.

2. Integrity: Integrity means that the data should not be changed or modified without permission. It ensures that the information remains accurate and trustworthy.

Example: A student's examination marks cannot be changed by unauthorized users in the university database.

3. Availability: Availability means that information and services should be available whenever authorized users need them. The system should work properly without unnecessary downtime.

Example: An ATM service should be available 24×7 so that customers can withdraw money at any time.

Advantages of CIA Triad

- Protects sensitive information from unauthorized access.
- Ensures data remains accurate and trustworthy.
- Keeps systems and services available whenever needed.

Q3. Compare Symmetric and Asymmetric Encryption Techniques with their Advantages, Disadvantages, and Applications.

- Encryption is the process of converting **plaintext** into **ciphertext** to protect data from unauthorized access. There are two main types of encryption:
 1. **Symmetric Encryption**
 2. **Asymmetric Encryption**

Symmetric Encryption

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In **Symmetric Encryption**, the **same secret key** is used for both encryption and decryption. The sender and receiver must share the same key securely.

Examples: DES, AES, Blowfish

Asymmetric Encryption

In **Asymmetric Encryption**, **two different keys** are used:

- **Public Key** – Used for encryption.
- **Private Key** – Used for decryption.

The public key can be shared with everyone, while the private key is kept secret.

Examples: RSA, ECC, ElGamal

Comparison between Symmetric and Asymmetric Encryption

Symmetric Encryption

Uses **one secret key**
Faster encryption
Less secure for key sharing
Suitable for large amounts of data
Example: DES, AES

Asymmetric Encryption

Uses **two keys** (Public & Private)
Slower encryption
More secure for key sharing
Suitable for small amounts of data
Example: RSA, ECC

Advantages

Symmetric Encryption

- Fast and efficient.
- Simple to implement.
- Suitable for encrypting large files.

Asymmetric Encryption

- Provides better security.
- Secure key exchange.
- Supports digital signatures and authentication.

Disadvantages

Symmetric Encryption

- Difficult to share the secret key securely.
- If the key is leaked, data security is lost.
- Does not provide digital signatures.

Asymmetric Encryption

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- Slower than symmetric encryption.
- Requires more processing power.
- More complex to implement.

Applications

Symmetric Encryption

- File encryption
- Disk encryption
- Database security

Asymmetric Encryption

- Digital signatures
- SSL/TLS (HTTPS websites)
- Secure email encryption

Q4. Explain the Working of the Caesar Cipher with a Suitable Example.

- The **Caesar Cipher** is one of the oldest and simplest encryption techniques in cryptography. It was developed by **Julius Caesar** to send secret military messages.
- In the Caesar Cipher, each letter of the plaintext is shifted by a fixed number of positions in the alphabet. This fixed number is called the **key (shift value)**.

For example, if the key is **3**, then:

- A → D
- B → E
- C → F

Working of Caesar Cipher

The encryption process follows these steps:

1. Choose a secret key (shift value).
2. Shift each letter of the plaintext by the key value.
3. The new letters form the **ciphertext**.
4. To decrypt, shift each ciphertext letter backward by the same key.

Example (Key = 3)

Plaintext

HELLO

Encryption



PPSU
P P SAVANI UNIVERSITY



NAAC
ACCREDITED



A+
GRADE

Shift +3	K	H	O	O	R
----------	---	---	---	---	---

KH00R

Shift -3	H	E	L	L	0
----------	---	---	---	---	---

HELLO

- Simple and easy to understand.
- Fast encryption and decryption.
- Useful for learning basic cryptography.

- Uses only one fixed key.
- Easy to break using brute-force attacks.
- Not secure for modern communication.

```
| Plaintext | --> | Encryption (+3) | --> | Ciphertext | --> | Decryption (-3) | --> | Plaintext |
```

- The **Vigenère Cipher** is a **polyalphabetic substitution cipher** that uses a **keyword** to encrypt the plaintext. Each letter of the plaintext is shifted according to the corresponding letter of the repeated keyword.

Plaintext: NETWORKSECURITY

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Keyword : P P S U P P S U P P S

Step 2: Convert Letters into Numbers

Use the following rule:

A = 0, B = 1, C = 2, ..., Z = 25

Plaintext	N	E	T	W	O	R	K	S	E	C	U	R	I	T	Y
Value	13	4	19	22	14	17	10	18	4	2	20	17	8	19	24
Keyword	P	P	S	U	P	P	S	U	P	P	S	U	P	P	S
Value	15	15	18	20	15	15	18	20	15	15	18	20	15	15	18

Step 3: Encryption Formula

Cipher = (Plaintext + Keyword) mod 26

P	K	(P+K) mod 26	Cipher
13	15	2	C
4	15	19	T
19	18	11	L
22	20	16	Q
14	15	3	D
17	15	6	G
10	18	2	C
18	20	12	M
4	15	19	T
2	15	17	R
20	18	12	M
17	20	11	L
8	15	23	X
19	15	8	I
24	18	16	Q

Step 4: Ciphertext

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Ciphertext = CTLQDGCMT RMLXIQ

Working Diagram

Plaintext + Keyword $\longrightarrow$ Encryption ($P + K \bmod 26$) $\longrightarrow$ Ciphertext

Advantages

- More secure than the Caesar Cipher.
- Uses a repeating keyword for encryption.
- Difficult to break without the correct keyword.

Q6. Encrypt the Plaintext "CRYPTOGRAPHY" using the Playfair Cipher with the Key "ICAPPSU". Show All Steps.

- The **Playfair Cipher** is a **digraph substitution cipher**, which encrypts **two letters (a pair)** at a time instead of one letter. It uses a **5×5 matrix** created from a secret key. In the matrix, **I and J are treated as the same letter**.

Given Plaintext: CRYPTOGRAPHY

Key: ICAPPSU

Step 1: Create the Playfair Matrix

Remove repeated letters from the key and fill the remaining alphabet (excluding J).

```
1 2 3 4 5
+---+---+---+---+
1 | I | C | A | P | S |
+---+---+---+---+
2 | U | B | D | E | F |
+---+---+---+---+
3 | G | H | K | L | M |
+---+---+---+---+
4 | N | O | Q | R | T |
+---+---+---+---+
5 | V | W | X | Y | Z |
+---+---+---+---+
```

Step 2: Divide Plaintext into Pairs

CR YP TO GR AP HY

Step 3: Encrypt Each Pair

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Pair	Rule Used	Cipher Pair
CR	Rectangle Rule	PY
YP	Rectangle Rule	SP
TO	Same Row	NQ
GR	Rectangle Rule	LN
AP	Same Row	PS
HY	Rectangle Rule	LW

Step 4: Ciphertext

Ciphertext = PYSPNQLNPSLW

Working Diagram:

Plaintext → Letter Pairs → Playfair Matrix → Encryption → Ciphertext

Advantages

- Encrypts two letters at a time, making it more secure than the Caesar Cipher.
- Difficult to break using simple frequency analysis.
- Easy to implement and understand.

Q7. Explain the DES Algorithm and the Feistel Structure with a Neat Block Diagram.

- **DES (Data Encryption Standard)** is a **symmetric-key encryption algorithm** used to secure digital data. It uses the **same secret key** for both encryption and decryption. DES encrypts data in **64-bit (8-byte) blocks** using a **56-bit secret key** and performs **16 rounds** of encryption based on the **Feistel Structure**.
- Although DES is now replaced by AES because of its smaller key size, it is still studied to understand the basics of cryptography.

Working of DES Algorithm

The working of DES consists of the following steps:

1. Plaintext (64 bits) is given as input.
2. An **Initial Permutation (IP)** is applied.
3. The data is divided into two equal parts:
 - Left Half (L)
 - Right Half (R)
4. The data passes through **16 Feistel Rounds** using the secret key.
5. After the 16th round, the left and right halves are combined.
6. Finally, the **Inverse Permutation (IP^{-1})** is applied to produce the ciphertext.

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Feistel Structure

The **Feistel Structure** is the main design used in DES. In each round:

- The **right half** is processed using a round key.
- The result is **XORed** with the left half.
- Then, the left and right halves are swapped.
- This process is repeated **16 times**.

Block Diagram of DES

Plaintext (64 bits)



Initial Permutation (IP)



Split into Left (L) and Right (R)



16 Feistel Rounds



Combine Left & Right



Inverse Permutation (IP^{-1})



Ciphertext (64 bits)

Simple Feistel Structure Diagram

Left (L) —┐

| XOR —————> New Right

Right (R) —> Function F + Round Key



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Swap L and R

Advantages

- Fast for encrypting small amounts of data.
- Simple and easy to implement.
- Uses the same key for encryption and decryption.

Limitations

- Uses only a **56-bit key**, which is weak today.
- Vulnerable to brute-force attacks.
- Replaced by AES for modern security.

Q8. Explain Shannon's Principles of Confusion and Diffusion. How are these Principles Implemented in Block Ciphers such as DES?

- **Claude Shannon** introduced two important principles of cryptography called **Confusion** and **Diffusion**. These principles help make encrypted data more secure and difficult for attackers to understand. Modern block ciphers such as **DES** use both principles to provide strong security.

1. Confusion

Confusion means hiding the relationship between the **secret key** and the **ciphertext**. It makes it difficult for an attacker to guess the secret key.

Example:

If the secret key changes, the ciphertext also changes completely, making it hard to predict the output.

2. Diffusion

Diffusion means spreading the information of the plaintext throughout the ciphertext. A small change in the plaintext causes many characters in the ciphertext to change.

Example:

Changing one letter in the message changes many letters in the encrypted output.

Implementation in DES

DES uses both **Confusion** and **Diffusion** in every round of encryption.

- **Confusion** is achieved using **Substitution (S-Boxes)**, which change the data based on the secret key.
- **Diffusion** is achieved using **Permutation (P-Boxes)**, which rearrange the bits so that the effect of one bit spreads throughout the ciphertext.

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These operations are repeated for **16 rounds**, making DES more secure.

Working Diagram

Plaintext → Confusion (S-Box) → Diffusion (P-Box) → Ciphertext

Advantages

- Increases the security of encrypted data.
- Makes cryptanalysis more difficult.
- Protects data from unauthorized access.

Q9. Compare the Caesar Cipher, Vigenère Cipher, and Playfair Cipher based on their Encryption Techniques, Key Usage, Advantages, Limitations, and Security Features. Support your answer with Suitable Examples.

- The **Caesar Cipher, Vigenère Cipher, and Playfair Cipher** are classical encryption techniques used to protect messages. They differ in the way they encrypt data, use keys, and provide security.

Feature	Caesar Cipher	Vigenère Cipher	Playfair Cipher
Encryption Technique	Shifts each letter by a fixed value	Uses a repeating keyword	Encrypts two letters (digraph) at a time
Key Used	One numeric key (e.g., 3)	Keyword (e.g., PPSU)	Keyword (e.g., ICAPPSU)
Example	HELLO → KHOOR	NETWORK → Ciphertext using keyword	CRYPTO → Ciphertext using Playfair matrix
Advantages	Simple and fast	More secure than Caesar	More secure than Caesar and Vigenère
Limitations	Easy to break	Keyword repetition can be attacked	Matrix creation is more complex
Security	Low	Medium	High (among classical ciphers)

Advantages of Classical Ciphers

- Easy to learn and implement.
- Useful for understanding cryptography concepts.
- Form the foundation of modern encryption techniques.

Q10. Explain the Hill Cipher Algorithm with a Suitable Example. Describe the Encryption and Decryption Process using Matrix Multiplication and Discuss its Advantages and Limitations.

The **Hill Cipher** is a **polyalphabetic substitution cipher** developed by **Lester S. Hill** in 1929. It uses **matrix multiplication** to encrypt and decrypt messages. Unlike the Caesar Cipher, it encrypts **a group of letters together**, making it more secure.

Encryption Process

1. Choose a **key matrix**.
2. Convert the plaintext letters into numbers using:
 - **A = 0, B = 1, C = 2, ... Z = 25**
3. Multiply the plaintext matrix by the key matrix.
4. Take the result **mod 26**.
5. Convert the numbers back into letters to get the ciphertext.

Example:

Plaintext

HI

Key Matrix

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Convert Letters into Numbers

H = 7

I = 8

Plaintext Matrix:

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Matrix Multiplication

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

Now take **mod 26**

$$45 \bmod 26 = 19$$

$$54 \bmod 26 = 2$$

Convert back into letters:

$$19 = T$$

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$2 = C$

Ciphertext

TC

Decryption Process

1. Find the **inverse of the key matrix**.
2. Multiply the ciphertext matrix by the inverse matrix.
3. Apply **mod 26**.
4. Convert the numbers back into letters.
5. The original plaintext is obtained.

Working Diagram

Plaintext $\longrightarrow$ Convert to Numbers $\longrightarrow$ Matrix Multiplication $\longrightarrow$ Mod 26 $\longrightarrow$ Ciphertext

Advantages

- More secure than the Caesar and Vigenère ciphers.
- Encrypts multiple letters together.
- Uses matrix mathematics, making encryption stronger.

Limitations

- Matrix calculations are difficult.
- The key matrix must have an inverse.
- Not suitable for modern high-security applications.

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